

HUDSON VALLEY, NY, USA

WMO: 725036

Lat: 41.626N

Lon: 73.882W

Elev: 51

StdP: 100.72

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14757

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.4	-13.4	-22.5	0.5	-13.8	-19.9	0.6	-11.3	9.6	-2.0	8.6	-2.4	1.5	10	0.468

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.9	32.9	23.1	31.3	22.4	29.9	21.6	24.7	30.4	23.9	29.0	23.1	27.7	3.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.9	17.7	27.4	22.3	17.1	26.8	21.5	16.3	25.9	74.9	30.4	71.6	28.9	68.7	27.9	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.3	7.5	6.4	-21.0	35.9	3.5	2.1	-23.5	37.4	-25.6	38.6	-27.6	39.8	-30.2	41.4		
(5)			-21.3	26.5	3.5	1.1	-23.8	27.3	-25.8	28.0	-27.7	28.7	-30.2	29.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.5	-2.8	-1.5	3.2	9.6	15.3	20.3	23.2	22.2	17.9	11.5	5.4	0.3	(6)
(7)		DBStd	10.05	5.96	5.35	5.18	4.61	4.18	3.53	2.96	2.99	3.85	4.34	4.69	4.93	(7)
(8)		HDD10.0	1492	397	322	219	61	4	0	0	0	1	34	151	303	(8)
(9)		HDD18.3	3298	654	554	469	264	114	20	2	4	54	216	387	559	(9)
(10)		CDD10.0	1657	1	1	8	49	169	308	409	377	238	80	14	2	(10)
(11)		CDD18.3	421	0	0	0	3	21	78	152	123	42	3	0	0	(11)
(12)		CDH23.3	4041	0	0	5	62	273	762	1466	1074	362	34	1	0	(12)
(13)		CDH26.7	1378	0	0	1	17	84	251	564	359	97	4	0	0	(13)
(14)	Wind	WSAvg	2.3	2.5	2.6	2.9	2.8	2.3	2.1	1.9	1.6	1.7	2.1	2.3	2.3	(14)
(15)	Precipitation	PrecAvg	1119	74	66	82	94	111	102	113	100	99	92	91	85	(15)
(16)		PrecMax	1554	211	144	197	227	314	224	367	294	320	462	218	233	(16)
(17)		PrecMin	661	12	9	4	28	25	8	19	17	12	10	18	15	(17)
(18)		PrecStd	215	42	29	37	40	63	54	72	57	63	76	47	46	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.1	16.5	23.0	29.0	32.0	33.9	35.9	34.2	32.2	26.9	21.3	17.4	(19)
(20)			MCWB	13.8	12.1	15.0	17.5	20.9	24.0	24.3	24.0	22.3	19.3	15.7	13.5	(20)
(21)		2%	DB	11.8	12.3	18.1	24.6	29.2	31.6	33.4	31.9	29.4	23.8	18.5	13.2	(21)
(22)			MCWB	9.5	7.7	11.7	15.1	19.7	22.4	23.4	23.1	21.2	17.7	14.6	10.7	(22)
(23)		5%	DB	7.7	9.0	14.7	21.3	26.6	29.5	31.6	30.2	27.3	21.7	16.2	10.7	(23)
(24)			MCWB	5.3	5.4	9.0	13.6	18.2	21.4	22.9	22.2	20.6	17.1	12.4	8.3	(24)
(25)		10%	DB	5.3	6.3	11.7	18.4	23.8	27.7	29.8	28.7	25.3	19.4	13.7	7.7	(25)
(26)			MCWB	2.9	3.1	7.6	11.7	16.8	20.4	21.9	21.4	19.5	15.1	10.3	5.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.3	13.5	16.1	19.1	22.9	25.1	26.1	25.8	24.3	21.2	17.7	14.6	(27)
(28)			MCD B	15.6	16.0	21.8	26.4	30.0	31.9	33.2	31.6	28.7	24.6	20.0	16.4	(28)
(29)		2%	WB	9.3	8.5	12.7	16.9	21.0	23.8	24.9	24.4	23.0	19.5	15.4	11.2	(29)
(30)			MCD B	10.7	11.1	16.8	22.5	26.6	29.4	31.0	29.4	27.1	22.3	17.8	12.9	(30)
(31)		5%	WB	5.6	5.6	10.1	14.8	19.6	22.8	24.0	23.6	21.8	17.7	13.2	8.2	(31)
(32)			MCD B	7.5	8.6	13.8	19.8	24.5	27.7	29.2	28.2	25.4	20.6	15.3	10.1	(32)
(33)		10%	WB	3.2	3.5	7.7	12.7	18.0	21.6	23.1	22.7	20.7	15.7	10.8	5.5	(33)
(34)			MCD B	5.0	6.0	11.2	17.1	22.3	26.0	27.9	26.8	24.0	18.7	13.2	7.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.4	10.3	10.8	12.3	12.4	11.9	11.9	11.7	12.2	11.7	10.6	8.8	(35)
(36)			MCD BR	11.6	14.0	15.7	17.5	16.4	14.3	13.7	13.3	14.0	14.6	13.6	11.7	(36)
(37)			MCWBR	9.5	10.5	9.9	9.3	8.1	6.7	5.5	5.6	7.1	9.0	9.9	10.0	(37)
(38)		5% WB	MCD BR	11.1	12.6	14.1	14.9	13.9	12.1	11.6	11.1	11.4	12.2	11.7	11.1	(38)
(39)			MCWBR	9.7	10.0	10.1	9.1	7.4	6.3	5.4	5.4	6.5	8.7	9.8	10.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.305	0.323	0.348	0.370	0.413	0.471	0.483	0.461	0.386	0.356	0.323	0.304	(40)	
(41)		taud	2.403	2.367	2.383	2.374	2.280	2.153	2.132	2.184	2.370	2.462	2.508	2.490	(41)	
(42)		Ebn at Noon	843	881	899	901	865	812	797	803	848	835	820	818	(42)	
(43)		Edh at Noon	80	97	107	117	131	150	151	140	108	88	73	69	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.77	2.65	3.65	4.64	5.19	5.64	5.86	5.15	4.24	2.83	1.96	1.45	(44)	
(45)		RadStd	0.13	0.25	0.28	0.50	0.52	0.52	0.34	0.32	0.42	0.30	0.18	0.15	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	-0.30	N/A	N/A	+114	+63
(47) Regional (61 neighbors)		+0.39	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+108	+58

Nomenclature: See separate page